Milestone 1 - STEAMingHot
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Dataset

We have chosen to create our own dataset for this project through scrapping, using a modified version of the Steam-Games-Scraper from FronkonGames. While pre-built datasets generated by this script are available on Kaggle, our variant will offer greater flexibility and allows us to include more recent data on successful releases, such as *Clair Obscur: Expedition 33*.

The data accuracy is very high because the script retrieves information from two highly reliable sources: the official Steam Web API for direct game data, and SteamSpy for other data. However, certain very specific events within the Steam platform and its community may introduce some bias. For instance:

- The "free weekend effect": some obscure free games are free for a weekend, and they are counted as "owned" for everyone in the database. (more detailed explanations)
- Review bombing: coordinated effort of an online community to sabotage a game. These are usually handled on the games public web pages, but probably not for raw data fetched from the APIs

We will only be able to catch those when our custom dataset will be built.

Problematic

Our visualization is intended to help game developers understand what makes a game become popular based purely on data and statistics. This will be achieved with multiple tools, for example:

- Game-specific statistics, such as:
    - a heatmap that shows how the target audience for a game was dispatched across the world
    - insights on text reviews, positive VS negative reviews, recommendations
    - peak-playing times depending on periods of the year or new game patches/versions.
- A bar chart race to visualize the most played games on Steam, across the years.
- A map of the world that gives country-specific statistics about games (most played games, categories, genres).
- A correlation between games' buying peaks and discounts or release date.
- A world map that allows the user to view general gaming tendencies in each country.

Exploratory Data Analysis

AppID	Name	Release date	Estimated owners	Price	About the game	Reviews	Metacritic score	Positive	Negative	Achievements	Recommendations	Developers
440	Team Fortress 2	10/10/2007	50000000 - 100000000	0	'The most fun you can have online' - PC Gamer Is now FREE! T		92	1044264	117208	520	45622	Valve
271590	Grand Theft Auto V Legacy	13/04/2015	50000000 - 100000000	0	When a young street hustler, a retired bank robber and a terri		96	1739980	230576	77	1854262	Rockstar North
1063730	New World: Aeternum	28/09/2021	50000000 - 100000000	59.99	Explore a thrilling, massive and beautiful world filled with da		0	196798	90080	133	253342	Amazon Games
304930	Unturned	07/07/2017	50000000 - 100000000	0	STAY UNTURNED You're one of the few not yet turned zombie.		0	506516	48852	88	7544	Smartly Dressed Games
1938090	Call of Duty®	27/10/2022	50000000 - 100000000	0	The Call of Duty® experience supports Call of Duty®: Black Op		0	419594	294520	157	402945	Treyarch,Raven Software,B
1599340	Lost Ark	11/02/2022	50000000 - 100000000	0	Embark on an odyssey for the Lost Ark in a vast, vibrant world		0	143481	58540	156	65217	Smilegate RPG
1623730	Palworld	18/01/2024	50000000 - 100000000	22.49	Q. What kind of game is this? A. In this game, you can peacef		0	358266	22443	57	343464	Pocketpair
2358720	Black Myth: Wukong	19/08/2024	50000000 - 100000000	59.99	Black Myth: Wukong is an action RPG rooted in Chinese myth		0	1111720	38378	81	863021	Game Science
550	Left 4 Dead 2	16/11/2009	50000000 - 100000000	1.99	Set in the zombie apocalypse, Left 4 Dead 2 (L4D2) is the highl		89	940221	23762	101	770930	Valve
236850	Europa Universalis IV	13/08/2013	5000000 - 10000000	4.99	Developed by Paradox Tinto, the fourth instalment of the awa	"A hallmark of excellence."	87	116432	18004	373	94068	Paradox Development Stud
394360	Hearts of Iron IV	06/06/2016	5000000 - 10000000	14.99	Take charge of history's greatest war machines in Hearts of Ir	"Hearts of Iron 4 wields cor	83	305168	36982	266	264373	Paradox Development Stud
1100600	Football Manager 2020	19/11/2019	5000000 - 10000000	0	Run your football club, your way. Every decision counts in Fo		84	28341	3401	97	17115	Sports Interactive
294100	RimWorld	17/10/2018	5000000 - 10000000	27.99	RimWorld is a sci-fi colony sim driven by an intelligent AI sto		87	206216	4301	0	194436	Ludeon Studios
363970	Clicker Heroes	13/05/2015	5000000 - 10000000	0	Ever wondered what one quadrillion damage per second feel	"106 hours and counting."	0	54989	6315	111	756	Playsaurus
250900	The Binding of Isaac: Rebirth	04/11/2014	5000000 - 10000000	7.49	When Isaac's mother starts hearing the voice of God demanc		0	334165	9255	641	337991	Nicalis, Inc.,Edmund McMi
582660	Black Desert	24/05/2017	5000000 - 10000000	0.99	Go Beyond Limits : WORLD CLASS MMORPG Black Desert is a g		80	64416	22503	0	60756	Pearl Abyss
427520	Factorio	14/08/2020	5000000 - 10000000	35	Factorio is a game in which you build and maintain factories		90	208770	6439	88	184924	Wube Software LTD.
1811260	EA SPORTS™ FIFA 23	30/09/2022	5000000 - 10000000	0	The World's Game Experience the pinnacle of women's inter		0	86192	65227	39	124136	EA Canada,EA Romania
594570	Total War: WARHAMMER II	28/09/2017	5000000 - 10000000	14.99	DEFEND YOUR WORLD. DESTROY THEIRS. Total War: WARHAM	"Four fun, distinct factions	87	114776	9476	152	87572	CREATIVE ASSEMBLY,Feral I
2195250	EA SPORTS FC™ 24	28/09/2023	5000000 - 10000000	0	This game includes optional in-game purchases of virtual cu		0	63281	46293	40	93950	EA Canada,EA Romania
767560	War Robots	04/04/2018	5000000 - 10000000	0	Great to see you, Commander! War Robots is the biggest sho		0	15951	10624	0	0	MY.GAMES
440900	Conan Exiles	08/05/2018	5000000 - 10000000	9.99	Conan Exiles is an online multiplayer survival game, now wit	"...the genre has saved one	0	84575	22890	36	74676	Funcom
273110	Counter-Strike Nexon	07/10/2014	5000000 - 10000000	0	About Counter-Strike Nexon Counter-Strike Nexon is a Free to		0	28373	14329	168	585	Valve,NEXON
306130	The Elder Scrolls® Online	22/05/2017	5000000 - 10000000	4.99	Every legend starts somewhere and in The Elder Scrolls Onlin		80	125497	28616	0	130859	ZeniMax Online Studios
386180	Crossout	26/07/2017	5000000 - 10000000	0	Crossout — is a postapocalyptic MMO action game where yo	"Like if Mad Max and Twiste	0	48219	15920	193	883	Targem Games
2669320	EA SPORTS FC™ 25	26/09/2024	5000000 - 10000000	69.99	Introducing Club Membership – the next phase of loyalty in EA		0	37253	37543	39	93317	EA Canada,EA Romania
72850	The Elder Scrolls V: Skyrim	10/11/2011	5000000 - 10000000	19.99	EPIC FANTASY REBORN The next chapter in the highly anticipa	"A Masterpiece" 10/10 – Eur	94	300915	16364	75	181105	Bethesda Game Studios
779340	Total War: THREE KINGDOMS	23/05/2019	5000000 - 10000000	11.99	Total War: THREE KINGDOMS is the first in the multi award-wi	"The example for all games	85	74951	26794	99	70794	CREATIVE ASSEMBLY,Feral I
1049590	Eternal Return	19/07/2023	5000000 - 10000000	0	Search, Craft, Hunt! The eternal experiment continues. On the		0	51694	13352	0	106	Nimble Neuron
365590	Tom Clancy's The Division™	07/03/2016	5000000 - 10000000	4.49	During Black Friday, a devastating pandemic sweeps throug		0	65652	27546	60	74755	Massive Entertainment
48700	Mount & Blade: Warband	31/03/2010	5000000 - 10000000	4.99	In a land torn asunder by incessant warfare, it is time to asse		78	162632	3773	80	132980	TaleWorlds Entertainment
1446780	MONSTER HUNTER RISE	12/01/2022	5000000 - 10000000	7.99	Rise to the challenge and join the hunt! In Monster Hunter Ris	"The greatest entry in Capc	86	92564	20108	100	74439	CAPCOM Co., Ltd.
322170	Geometry Dash	22/12/2014	5000000 - 10000000	2.49	Jump and fly your way through danger in this rhythm-based a	"Frustratingly wonderful." K	0	401026	28895	547	457790	RobTop Games
10190	Call of Duty®: Modern Warfare® 2 (2009)	11/11/2009	5000000 - 10000000	14.99	The most-anticipated game of the year and the sequel to the I		86	31945	2927	50	34262	Infinity Ward
513710	SCUM	17/06/2025	5000000 - 10000000	17.99	Welcome prisoners to SCUM Island! Where participation is n	"Scum has a rare crafting s	0	83704	28411	69	93844	Gamepires
766570	Russian Fishing 4	10/11/2021	5000000 - 10000000	0	Experience more than 230 detailed species. Over time, this nu		0	39598	33887	145	0	FishSoft LLC
489830	The Elder Scrolls V: Skyrim Special Editio	27/10/2016	5000000 - 10000000	9.99	Winner of more than 200 Game of the Year Awards, The Elder		74	310162	26896	75	201604	Bethesda Game Studios

As we have not built our custom dataset yet, we rely on [this dataset](), resulting from the *Steam Games Scraper* script mentioned above, to present a small preview of the data in the screenshot above (it's tiny sorry). The image shows the first few most interesting columns of data that we can get for each game (the games have been sorted by Most Owned). There are over 40 columns of data for every game on the Steam platform.
We intend to fetch the most relevant data for our website by excluding columns that we won't be using in our visualizations (required age, movies, support email address, …). We will also fetch the first couple hundred most popular/recent/played games in order to filter out smaller irrelevant games.

Related work

Based on a paper published by Hu and co. [1], the researchers used a dataset similar to ours and were able to collect and use multiple Machine Learning techniques to predict the popularity of Steam games. Their approach was to analyze the data to predict the favourability of the reviews given to the games. They correlated the amount of platforms the games were compatible on, and their popularity and ratings.

For this project, we plan to use easy-to-understand graphics and novel correlation pairings in order to better pinpoint the underlying causes of a game's popularity, differing from more surface-level analysis. We also want to make our data visualization website about video games as playful as possible; after all, we're gamers before we are developers.
Our project is inspired by SteamDB, a huge database of Steam's games inventory and filled with diverse information on each game such as genres, price history, ratings history, peak number of concurrent users (CCU) and much more.
We will also take some inspiration on various types of graph animations such as bar chart races and build our own with the dataset.

References

[1] Hu, W., Wang, Y. and Xia, R. Machine Learning-Based Steam Platform Game's Popularity Analysis, 2024. DOI: 10.5220/0012853800004547